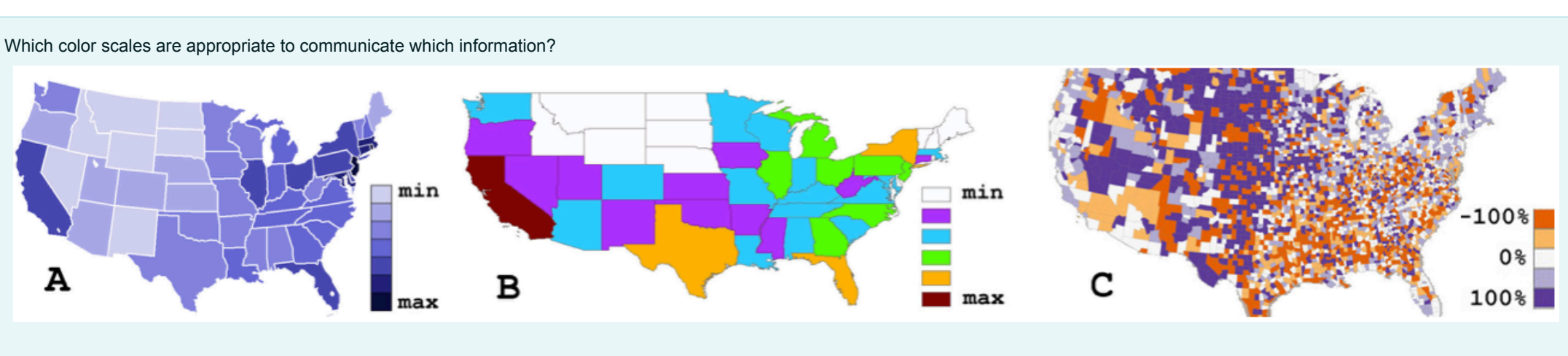
Frage 1
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Mark the **correct** statements about geospatial data

Wählen Sie eine oder mehrere Antworten:
- ☑ **Surface phenomena** can be specified by a series of longitude, latitude and height.
- ☐ **Point phenomena** are termed one-dimensional
- ☐ **Surface phenomena** are termed three-dimensional.
- ☐ Examples of **area phenomena** are cities, and streets.

Frage 2
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Which color scales are appropriate to communicate which information?



Wählen Sie eine oder mehrere Antworten:
- ☑ Map C is best suited to communicate the bettering or worsening of average income per county since last year
- ☐ Map B is best suited to communicate population per square mile per state
- ☐ Map A is best suited to communicate the bettering or worsening of average income per state since last year
- ☑ Map A is best suited to communicate population per square mile per state

Frage 3
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

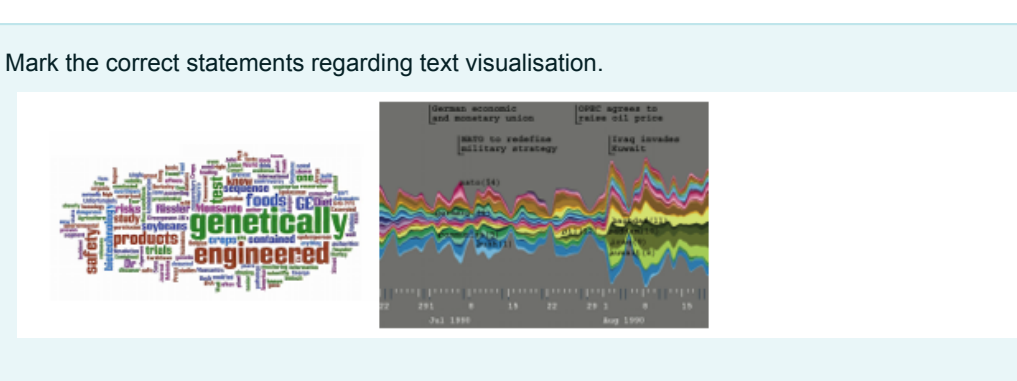
Map projections are concerned with mapping the positions on the globe (sphere) to positions on the screen (flat surface).

Mark the correct statements.

Wählen Sie eine oder mehrere Antworten:
- ☐ **Azimuthal projections** do not preserve the direction from a central point.
- ☑ In **cone projections** the degrees of longitude and latitude are usually orthogonal to each other.
- ☐ **Conformal projections** do not retain local angles on each point of a map correctly.
- ☑ **Conformal projection** does preserve shapes.

Frage 4
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

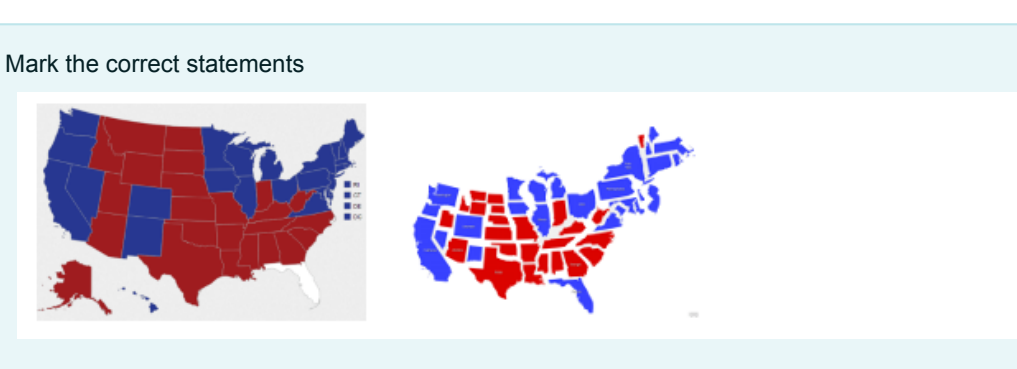
Mark the correct statements regarding text visualisation.



Wählen Sie eine oder mehrere Antworten:
- ☑ **ThemeRiver** (stream graph) is a visualization of thematic changes in a document collection over time.
- ☐ The **semantic level** of text representation deals with identifying and tagging each token's function inside the sentence.
- ☑ Text visualization is useful to compare patterns in text documents.
- ☐ **Word clouds** are visualizations of term frequencies as well as their context.

Frage 5
Vollständig
Erreichte Punkte 2,00 von 3,00
Frage markieren

Mark the correct statements



Wählen Sie eine oder mehrere Antworten:
- ☑ **Choropleth maps** communicate the geographical distribution of the represented statistical value **better than cartograms**.
- ☐ **Cartograms** are based on the assumption that the mapped attribute is uniformly distributed in the geographical region.
- ☑ **Cartograms** distort the geography according to the displayed statistical value.

Frage 6
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Concerning the aesthetics for dynamic visualization of graphs, mark the true sentences.

Wählen Sie eine oder mehrere Antworten:
- ☐ a. Concerning scalability, it only should be addressed for vertices and edges.
- ☐ b. "Temporal aliases" do not impact significantly the quality of the layout.
- ☑ c. "Reducing cognitive load" means reducing the amount of information the user has to keep in mind during the visualization of the graph.
- ☑ d. Is preferable to preserve the mental map in the design of a dynamic network visualization method.

Frage 7
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

A force-directed layout algorithm

Wählen Sie eine oder mehrere Antworten:
- ☑ a. models the graph after a physical system
- ☐ b. is a method to reorder rows and columns of a matrix
- ☑ c. is a graph-drawing method for node-link diagrams
- ☐ d. is used to compute the colours of a heatmap
- ☐ e. includes an energy maximization algorithm

Frage 8
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

When designing a node-link graph visualization, particular attention has to be put on how to visualize the edges. Mark the correct sentences.

Wählen Sie eine oder mehrere Antworten:
- ☑ a. Polyline edges with multiple bends per edge can be used, but the number of bends should be minimized.
- ☐ b. Curved approaches, such as Lombardi drawings, are usually preferred by the users even if they do not provide an advantage in any observed task.
- ☑ c. In the directed case, when arrows are used to indicate the direction, they should not overlap with other edges or arrow heads.
- ☐ d. Straight line node-link representations allow bends to reduce the number of crossings in the layout.

Frage 9
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

A treemap

Wählen Sie eine oder mehrere Antworten:
- ☐ a. is a non-space-filling visualization method
- ☑ b. is used to visualize hierarchies
- ☑ c. can be squarified
- ☑ d. can be nested
- ☐ e. is used to visualize vegetation in a geographic area

Frage 10
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Which of the following visual encodings for hierarchies satisfy the following description?

Features:
- Hierarchy visualized via parent-child relationship
- Layout techniques include "Layered" and "Radial" approaches
- No cycles

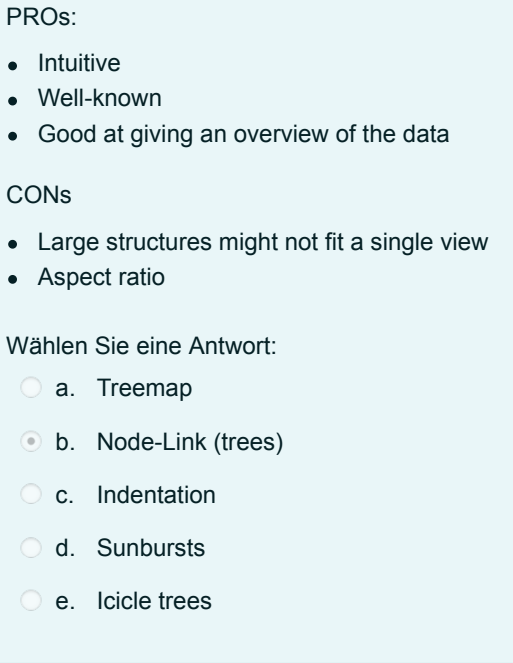
PROs:
- Intuitive
- Well-known
- Good at giving an overview of the data

CONs
- Large structures might not fit a single view
- Aspect ratio

Wählen Sie eine Antwort:
- ○ a. Treemap
- ● b. Node-Link (trees)
- ○ c. Indentation
- ○ d. Sunbursts
- ○ e. Icicle trees

Frage 11
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Semantic zooming ...



Wählen Sie eine oder mehrere Antworten:
- ☐ a. shows different visual representations depending on the color-coding chosen by the user.
- ☐ b. relies solely on geometric zooming.
- ☑ c. shows different visual representations depending on the zoom level chosen by the user and the available space for the visualization.
- ☑ d. allows users to choose between a large scope (i.e., zooming out and seeing an overview of the big picture) or a more detailed view (i.e., zooming in and seing all details about a limited range of the data).

Frage 12
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

How does the use of Focus+Context techniques impact usability?

Wählen Sie eine oder mehrere Antworten:
- ☑ a. overview+detail and fisheye interfaces are beneficial for specific tasks
- ☐ b. overview+detail and zooming interfaces impact users' performance negatively
- ☐ c. Focus+Context techniques are easy to follow and can be used without prior knowledge
- ☑ d. the effectiveness of Focus+Context techniques depends on the users' tasks and goals

Frage 13
Vollständig
Erreichte Punkte 0,00 von 3,00
Frage markieren

What levels of the Info-Vis Reference Model are influenced by dynamic querying (with real time feedback)?

Wählen Sie eine oder mehrere Antworten:
- ☐ a. Visual Mappings
- ☑ b. Visual Structures
- ☐ c. Data Transformations
- ☐ d. Data Tables
- ☑ e. Views
- ☐ f. View Transformations
- ☐ g. Raw Data

Frage 14
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Interaction modeling is employed to ...

Wählen Sie eine oder mehrere Antworten:
- ☑ a. ... validate effectiveness and user guidance of different interaction approaches.
- ☑ b. ... capture requirements the system can be developed and evaluated against.
- ☐ c. ... demonstrate interactions and tasks to users.

Frage 15
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Coordinated Multiple View systems ...

Wählen Sie eine oder mehrere Antworten:
- ☑ a. use two or more distinct views to support the investigation of a single conceptual entity
- ☑ b. combine different visualization techniques and apply them on different data representations
- ☐ c. use color-coding for trend detection
- ☐ d. can not be supported by brushing & linking techniques
- ☐ e. always support semantic zooming
- ☑ f. are often supported by brushing & linking techniques

Frage 16
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Tamara Munzner: The nested model of visualization design and validation captures

Wählen Sie eine oder mehrere Antworten:
- ☑ a. data/task abstraction
- ☑ b. domain situation
- ☐ c. patterns characteristics
- ☐ d. search functionality
- ☑ e. algorithm
- ☑ f. visual encoding/interaction idiom
- ☐ g. images properties

Frage 17
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

InfoVis Reference Model: User interaction can feed back into ...

Wählen Sie eine oder mehrere Antworten:
- ☑ a. data transformation
- ☐ b. table structure
- ☑ c. visual mapping
- ☐ d. text generation
- ☐ e. image manipulation
- ☑ f. view transformation

Frage 18
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

The first action in the "Knowledge Crystallization Loop" is ...

Wählen Sie eine Antwort:
- ○ a. ... archive huge amount of data
- ● b. ... forage for data
- ○ c. ... visualize data

Frage 19
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

The visualization design triangle [Miksch & Aigner 2014] specifies ...

Wählen Sie eine oder mehrere Antworten:
- ☑ a. user
- ☑ b. data
- ☐ c. software
- ☑ d. task
- ☐ e. environment
- ☐ f. hardware

Frage 20
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Anscombe's Quartett illustrates

Wählen Sie eine Antwort:
- ○ a. random variables, different graphs
- ○ b. high level goals, correct measures
- ● c. same statistics, different graphs

Frage 21
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

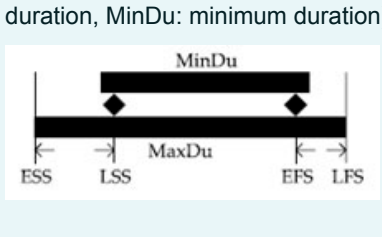
Chernoff Faces

Wählen Sie eine oder mehrere Antworten:
- ☐ a. are suited to communicate complex data effectively
- ☐ b. their features are likely to be perceived equally strong
- ☑ c. are suspected to convey emotions
- ☑ d. encode multiple data values with different graphical features

Frage 22
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Glyph or icon based techniques

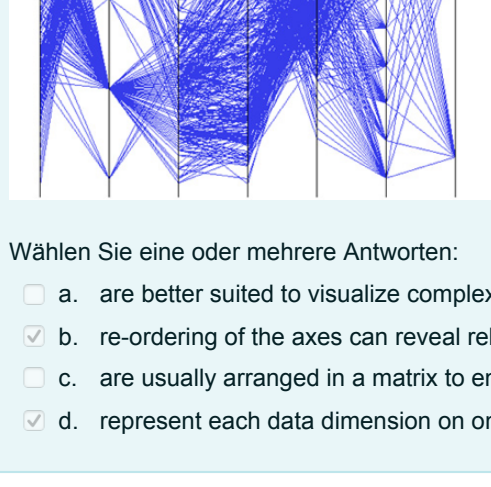
(Figure: example of a glyph: The *Time Annotation Glyph* represents temporal constraints of a medical treatment plan – ESS: earliest starting shift, LSS: latest starting shift, EFS: earliest finishing shift, LFS: latest finishing shift, MaxDu: maximum duration, MinDu: minimum duration)



Wählen Sie eine oder mehrere Antworten:
- ☑ a. are composed of graphical primitives
- ☐ b. cannot represent multiple data values at once
- ☑ c. do not represent one meaning as a whole but encode multiple data values with different graphical features

Frage 23
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

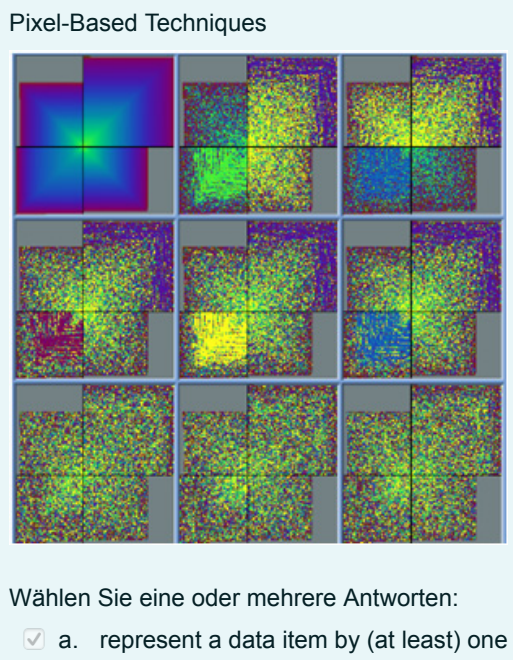
Parallel Coordinates



Wählen Sie eine oder mehrere Antworten:
- ☐ a. are better suited to visualize complex relations between two dimensions than scatterplots
- ☑ b. re-ordering of the axes can reveal relations between two data dimensions
- ☐ c. are usually arranged in a matrix to enable the visualization of multiple dimensions
- ☑ d. represent each data dimension on one of multiple parallel axes

Frage 24
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

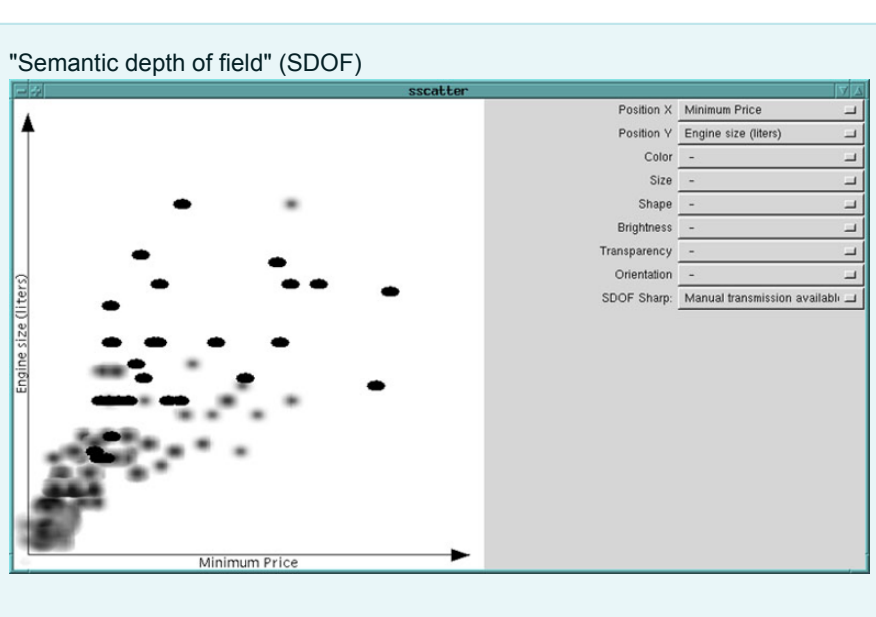
Pixel-Based Techniques



Wählen Sie eine oder mehrere Antworten:
- ☑ a. represent a data item by (at least) one pixel
- ☐ b. are suited for low-dimensional data sets only
- ☑ c. value ranges of attributes are mapped to a color map

Frage 25
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

"Semantic depth of field" (SDOF)



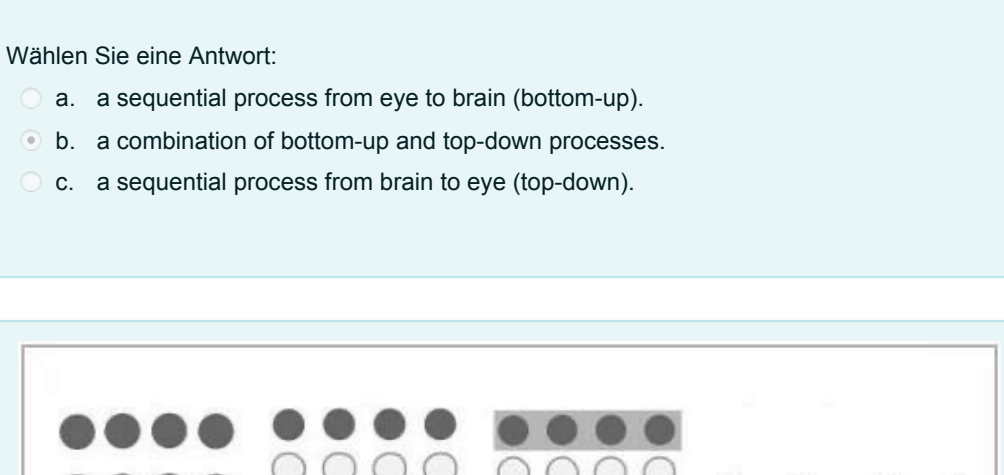
Wählen Sie eine oder mehrere Antworten:
- ☐ a. is good at visually communicating lots of different quantitative values
- ☑ b. can be used to visually represent an additional data dimension in scatterplots
- ☑ c. blurs data items based on their relevance

Frage 26
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Visual perception is...

Wählen Sie eine Antwort:
- ○ a. a sequential process from eye to brain (bottom-up).
- ● b. a combination of bottom-up and top-down processes.
- ○ c. a sequential process from brain to eye (top-down).

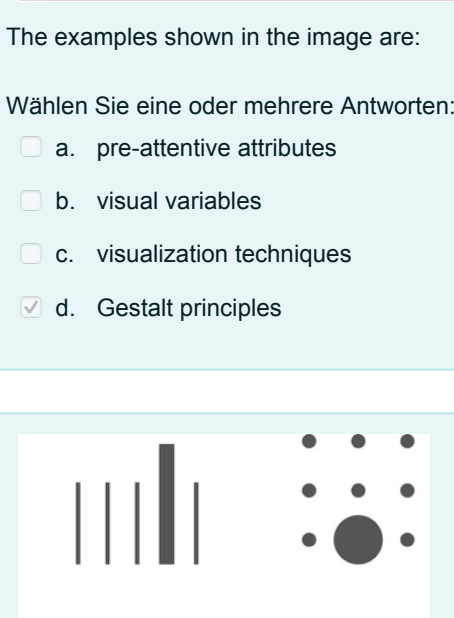
Frage 27
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren



The examples shown in the image are:

Wählen Sie eine oder mehrere Antworten:
- ☐ a. pre-attentive attributes
- ☐ b. visual variables
- ☐ c. visualization techniques
- ☑ d. Gestalt principles

Frage 28
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren



The examples shown in the image are:

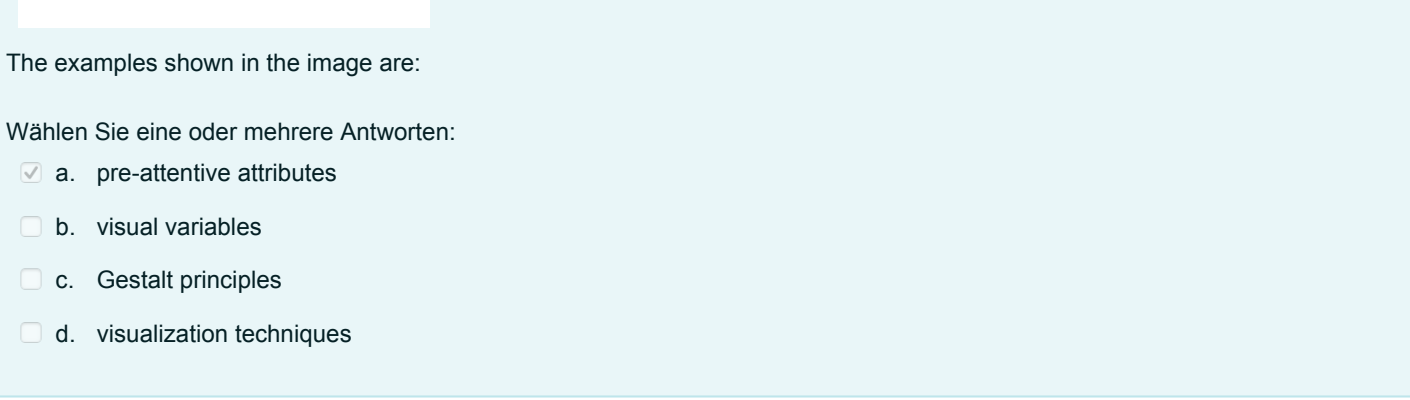
Wählen Sie eine oder mehrere Antworten:
- ☑ a. pre-attentive attributes
- ☐ b. visual variables
- ☐ c. Gestalt principles
- ☐ d. visualization techniques

Frage 29
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Along which scale are T-Shirt sizes measured?

Wählen Sie eine Antwort:
- ● a. ordinal
- ○ b. quantitative - interval
- ○ c. nominal
- ○ d. quantitative - ratio

Frage 30
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren



Which of the following statements are correct?

Wählen Sie eine Antwort:
- ● a. a is not expressive, b is expressive
- ○ b. a and b are both expressive
- ○ c. a is equally expressive than b, but b is more effective

Frage 31
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Which of the following statements about mapping time to space are correct?

Wählen Sie eine oder mehrere Antworten:
- ☑ a. mapping of time to visual features
- ☐ b. well suited for following trends and movements
- ☑ c. direct comparison of parameters between different points in time is possible
- ☐ d. not well suited for analytic and explorative tasks

Frage 32
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Which of the following statements about the visual mapping of time are correct

Wählen Sie eine oder mehrere Antworten:
- ☑ a. animation (time to time) mapping supports to follow trends and movements
- ☐ b. animation (time to time) mapping is the simplest mapping
- ☑ c. static (time to space) mapping is well suited for explorative tasks
- ☑ d. static (time to space) mapping support direct comparison
- ☐ e. animation (time to time) mapping is well suited for explorative tasks

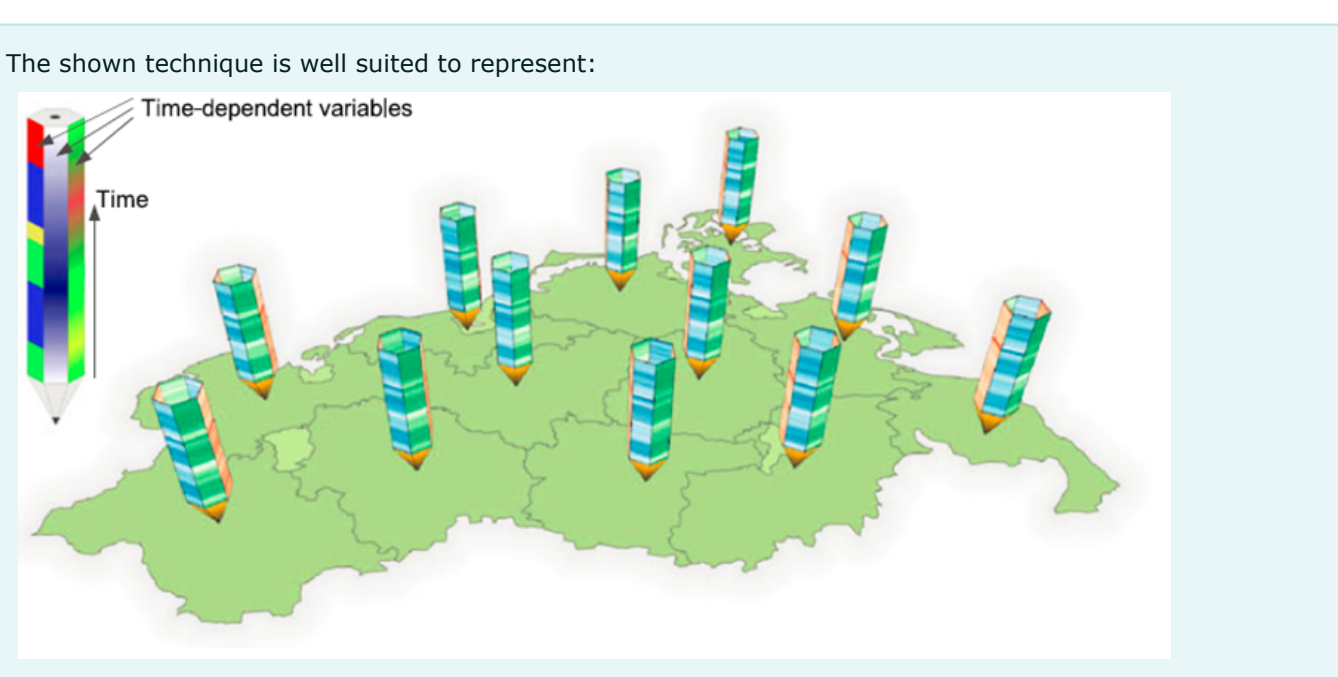
Frage 33
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

For modelling time-oriented data, possible viewpoints are:

Wählen Sie eine oder mehrere Antworten:
- ☑ a. branching
- ☐ b. cyclic
- ☑ c. multiple perspectives
- ☑ d. ordered
- ☐ e. interval-based

Frage 34
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

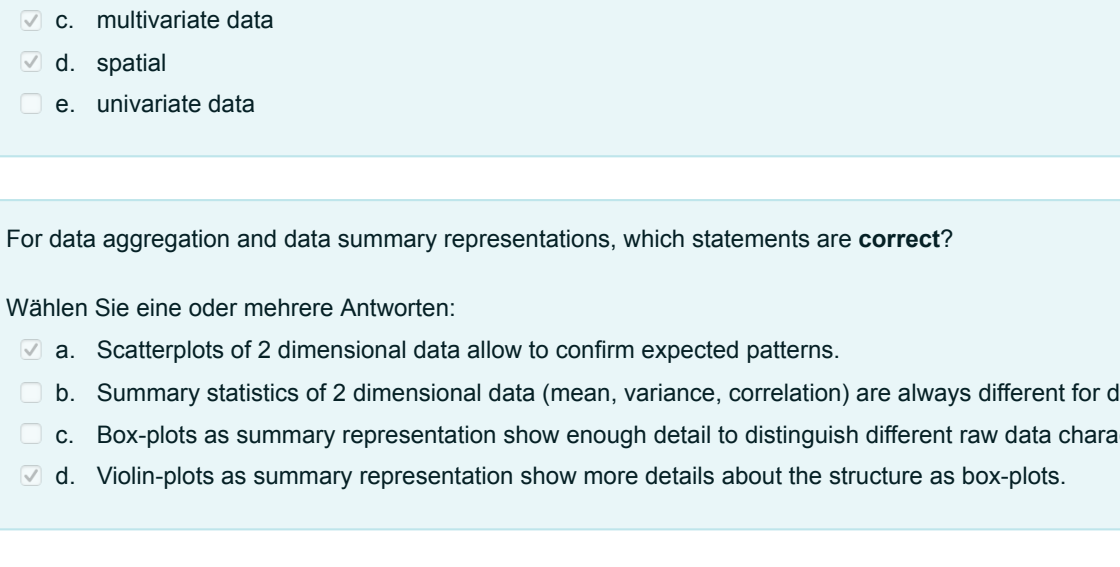
The shown techniques are well suited to represent:



Wählen Sie eine oder mehrere Antworten:
- ☑ a. univariate data
- ☐ b. multivariate data
- ☐ c. cyclic time
- ☑ d. time instant
- ☑ e. linear time

Frage 35
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

The shown technique is well suited to represent:



Wählen Sie eine oder mehrere Antworten:
- ☑ a. time instant
- ☐ b. cyclic time
- ☑ c. multivariate data
- ☑ d. spatial
- ☐ e. univariate data

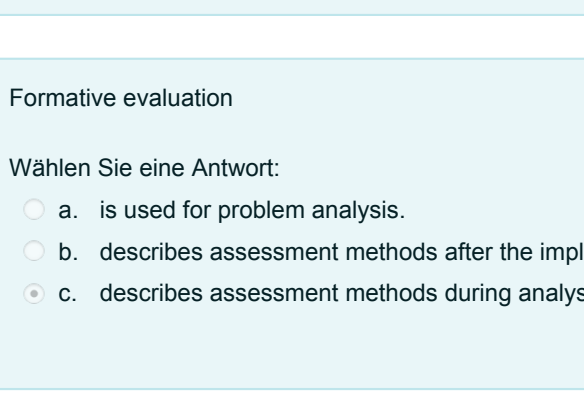
Frage 36
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

For data aggregation and data summary representations, which statements are **correct**?

Wählen Sie eine oder mehrere Antworten:
- ☑ a. Scatterplots of 2 dimensional data allow to confirm expected patterns.
- ☐ b. Summary statistics of 2 dimensional data (mean, variance, correlation) are always different for different datasets.
- ☐ c. Box-plots as summary representation show enough detail to distinguish different raw data characteristics.
- ☑ d. Violin-plots as summary representation show more details about the structure as box-plots.

Frage 37
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

According to the design study methodology (Sedlmair et al. 2012), when does a visualization design study offer a good solution for a given problem?



Wählen Sie eine Antwort:
- ○ a. Area "X": all the required information is in the head of the user
- ● b. Area "Y": either some of the required information is in the head of the user, or the task is not defined in a very clear way, or both.
- ○ c. Area "Z": the required information is available to the computer and the task is defined in a clear way.

Frage 38
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Formative evaluation

Wählen Sie eine Antwort:
- ○ a. is used for problem analysis.
- ○ b. describes assessment methods after the implementation phase.
- ● c. describes assessment methods during analysis and design phases.

Frage 39
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

The central questions of problem analysis are:

Wählen Sie eine oder mehrere Antworten:
- ☑ a. What kind of data are the users working with?
- ☐ b. What will the development method for the implementation phase be?
- ☑ c. What are the tasks of the users?
- ☐ d. How many software developers and designers are needed?
- ☑ e. Who are the users of the system?

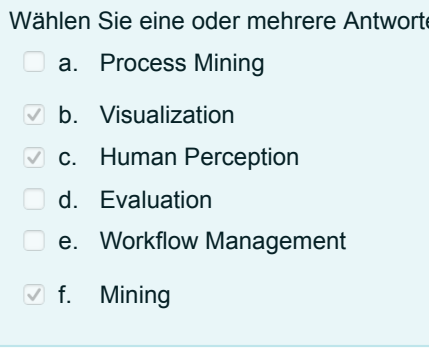
Frage 40
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

What are the main benefits of user-centered design/human-centered aspects?

Wählen Sie eine oder mehrere Antworten:
- ☐ a. systems get more complex and the discomfort is increased.
- ☐ b. product quality, aesthetics, and impact are reduced.
- ☑ c. productivity of users is usually improved.
- ☑ d. training and support costs are reduced.

Frage 41
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Components of Visual Analytics: Assign the correct terms to A, B, and C (in red letters)



Wählen Sie eine oder mehrere Antworten:
- ☐ a. Process Mining
- ☑ b. Visualization
- ☑ c. Human Perception
- ☐ d. Evaluation
- ☐ e. Workflow Management
- ☑ f. Mining

Frage 42
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Selected goals of Visual Analytics are designing and creating methods to enable users to ...

Wählen Sie eine oder mehrere Antworten:
- ☑ a. detect the expected and discover the unexpected
- ☐ b. render realistic images in an expressive way
- ☑ c. synthesize information and derive insight from massive, dynamic, ambiguous, and often conflicting data

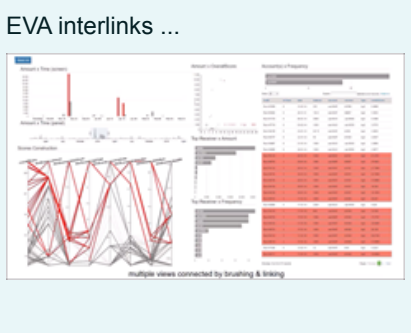
Frage 43
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Guidance is ...

Wählen Sie eine Antwort:
- ○ a. a computer-assisted process that aims to harmonize heterogeneous information sources
- ○ b. a computer-assisted process that aims to ease data quality exploration
- ● c. a computer-assisted process that aims to actively resolve a knowledge gap encountered by users during an interactive Visual Analytics session

Frage 44
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

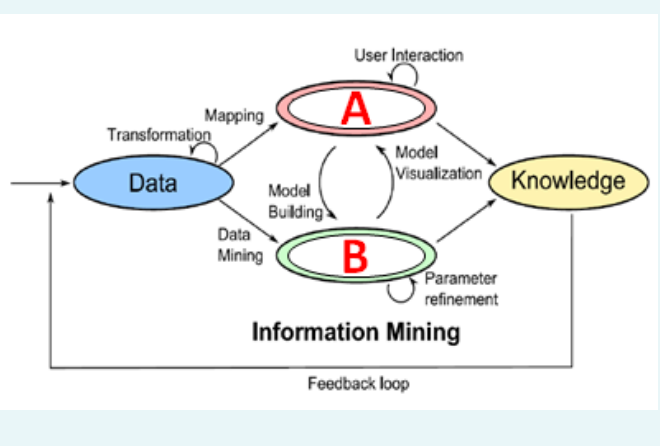
EVA interlinks ...



Wählen Sie eine Antwort:
- ○ a. heatmaps, parallel coordinates, scatter plots, networks, maps, and time lines
- ○ b. maps, line chart, parallel coordinates, scatter plot, row charts, and networks
- ● c. bar chart, line chart, parallel coordinates, scatter plot, row charts, and a table

Frage 45
Vollständig
Erreichte Punkte 3,00 von 3,00
Frage markieren

Visual Analytics – Process [Keim, et al. 2008]: Assign the correct terms to A and B (in red letters)



Wählen Sie eine oder mehrere Antworten:
- ☐ a. Information Retrieval
- ☑ b. Model
- ☑ c. Visualization
- ☐ d. Machine Learning